

- BC 8. If $y = \sin x$ and $y^{(n)}$ means “the n th derivative of y with respect to x ,” then the smallest positive integer n for which $y^{(n)} = y$ is

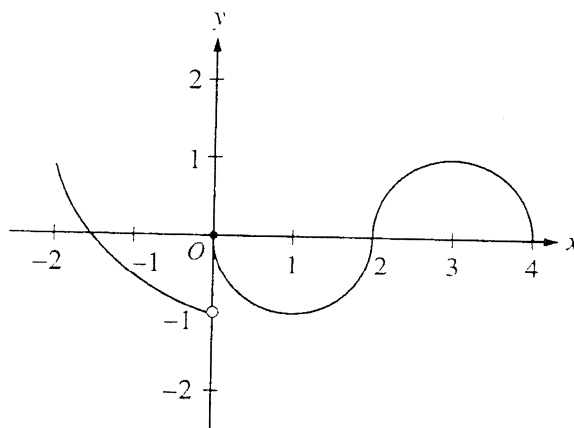
(A) 2 (B) 4 (C) 5 (D) 6 (E) 8

- BC 10. $\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$ is

(A) 0 (B) 1 (C) $\sin x$ (D) $\cos x$ (E) nonexistent

- BC 6. If $f(x) = \frac{x}{\tan x}$, then $f'\left(\frac{\pi}{4}\right) =$

(A) 2 (B) $\frac{1}{2}$ (C) $1 + \frac{\pi}{2}$ (D) $\frac{\pi}{2} - 1$ (E) $1 - \frac{\pi}{2}$



- BC 13. The graph of the function f shown in the figure above has a vertical tangent at the point $(2, 0)$ and horizontal tangents at the points $(1, -1)$ and $(3, 1)$. For what values of x , $-2 < x < 4$, is f not differentiable?

(A) 0 only (B) 0 and 2 only (C) 1 and 3 only (D) 0, 1, and 3 only (E) 0, 1, 2, and 3